Shubham Raj Sinha

Sonebhadra, Uttar Pradesh | shubhamrajsinha777@gmail.com | +91-9335269313
linkedin.com/in/shubham-raj-sinha-940a4b35a | github.com/shubhamrajsinha11

Professional Summary

B.Tech CSE fresher (May 2026) specializing in Data Science — skilled in Python, SQL, and ML model development. Built and deployed two end-to-end data projects with real metrics: an ML healthcare app (83.7% accuracy, 246K+ records) and an environment-disease correlation dashboard (37.3M+ cases, Pearson r = 0.628). Seeking an entry-level Data Analyst or Data Science role.

Technical Skills

Languages: Python, SQL, HTML5, CSS3, JavaScript **Visualization:** Matplotlib, Seaborn, Plotly, Power BI, Tableau

ML & Statistics: Scikit-learn, Logistic Regression, Decision Trees, Random Forest, Feature Engineering, SelectKBest, PCA, Cross-Validation, ROC-AUC, F1-Score

Data Analysis: Pandas, NumPy, EDA, Hypothesis Testing, Outlier Detection, MS Excel (Pivot Tables, VLOOKUP) **Tools:** Git, GitHub, Streamlit, Jupyter, Google Colab, MySQL, MS SQL Server, SQLite

Internship & Training Experience

Data Science Trainee – QSpiders Software Testing & Training Institute Feb 2026 – Present

- Preprocessed 246,945-row tabular data (378 columns, 755 classes) using Pandas; trained and benchmarked supervised ML models (Logistic Regression, Decision Trees, Random Forest) with stratified cross-validation and ROC-AUC evaluation.
- Reduced 377 features to 200 via SelectKBest mutual-information selection, then applied PCA (180 components, 95% variance); executed outlier detection and StandardScaler normalization pipelines.

Data Analyst Trainee – Coder Roots Jun 2025 – Jul 2025

- Cleaned and transformed structured business data using SQL; conducted EDA to surface patterns, trends, correlations, and outliers across multiple domains.
- Built automated transformation workflows and interactive Power BI and Tableau dashboards for KPI monitoring; reduced report generation time by 75%.

Web Development Trainee – Smart Future Point Jun 2024 – Jul 2024

- Engineered responsive web pages using HTML5, CSS3, and JavaScript; delivered a Library Book Record System and Portfolio Website covering the full front-end development lifecycle.

Projects

CareSync – Healthcare Diagnosis & Recommendation Application 2024 – 2025

Python · Streamlit · Scikit-learn · Pandas · NumPy · Plotly · SQLite · Git

- End-to-end ML healthcare web app predicting **755 disease classes** from 133 symptoms; trained on **246,945 records** with Logistic Regression (auto-selected from 3 competing models), achieving **83.7% test accuracy** and **83.9% weighted F1-score**.
- Full ML pipeline: data ingestion → SelectKBest (377 → 200 features) → StandardScaler → PCA (180 components, 95% variance) → Streamlit deployment; integrated a **53,766-entry drug database** covering 2,637 drugs across 708 conditions for medicine recommendations.
- Modular architecture (app.py, auth.py, database.py, history.py, profile.py) with PBKDF2-SHA256 authentication, SQLite persistence (4 tables), and interactive Plotly health analytics dashboards.

Air Pollution & Respiratory Disease Correlation Analysis 2024

Python · Streamlit · Pandas · NumPy · Matplotlib · Seaborn

- Analyzed **14,400 records** spanning **8 years (2018–2025)** across **10 cities and 5 states** in North India; investigated environmental correlation with 3 diseases (Asthma, Dengue, Heat Stroke) covering **37.3M+ population-affected cases**.
- Conducted statistical hypothesis testing (Pearson $r = 0.628$, $p < 0.001$, $R^2 = 0.394$ between population exposure and mortality); implemented EPA-standard AQI breakpoints for **6 pollutants** (PM2.5, PM10, O, NO, SO, CO) with real-time OpenWeatherMap API integration.
- Built interactive Streamlit dashboard featuring a multifactor disease risk scoring engine (Dengue, Asthma, Heat Stroke), live AQI gauge, correlation heatmaps, and city-wise trend visualizations; modularized with caching and API error handling.

Education

Sardar Beant Singh State University, Gurdaspur B.Tech in Computer Science & Engineering Aug 2022 – May 2026

- CGPA: 7.8/10
- **Relevant Coursework:** Statistics & Probability, Database Management Systems, Machine Learning, Data Visualization, Python for Data Analysis, Data Structures & Algorithms